Do Hang Seng Index Daily Returns Follow a Normal Distribution?

An empirical test with option pricing check

Fariha Tasnim

October 2026

Code and Data: https://github.com/farihatasnim04/hang_seng_returns_analysis.git

Summary

I downloaded 2,654 daily closing prices of the Hang Seng Index, covering 2016-01-04 to 2026-10-16 and converted them into daily log returns. I then tested whether those returns follow a normal (bell-curve) distribution, which many standard finance models assume.

The returns did not follow that assumption. They have an excess kurtosis of approximately 7.15 and a skewness of -0.37. There were 24 days more than three standard deviations from the mean, against about 7 expected under normality, also 10 days more than four standard deviations away, against about 0.17 expected. A Jarque–Bera test rejects normality (statistic 5,718, $p < 0.001$).

In the second part, I priced a one year at- the-money call option on the index with Monte Carlo simulation and checked it against the Black–Scholes formula. With 1,000,000 simulated paths the two prices agreed to within 0.03% (2,451.35 against 2,450.58 index points). Both methods assume normally distributed returns, which Part 1 contradicts, these should be read as model prices rather than market prices.

1. Question and motivation

Many classic models in finance, including the Black–Scholes option-pricing formula, assume that daily changes in asset prices follow a log-normal distribution. In a normal distribution, extreme values are extremely rare. If real markets produce extreme days far more often than that, models built on the assumption will understate risk.

This project asks a simple question: **are the daily returns of the Hang Seng Index consistent with a normal distribution?** It then builds the standard option-pricing method that relies on that assumption, to see how the two fit together.

2. Data

- **Index:** Hang Seng Index (ticker ^HSI), Hong Kong's main stock market index.
- **Source:** Yahoo Finance, downloaded with the Python package yfinance.
- **Period:** 2016-01-04 to 2026-10-16 (download requested for January 2016 to October 2026).
- **Observations:** 2,654 trading days of closing prices, giving 2,653 daily returns.

3. Method (Part 1: testing normality)

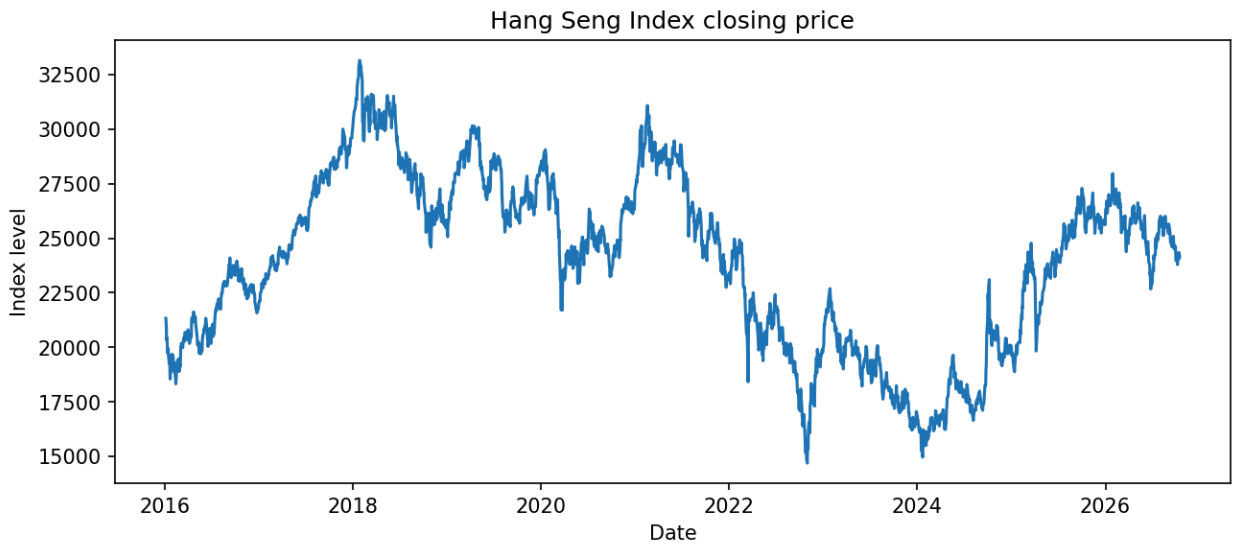
1. **Log returns.** For each day I computed $r(t) = \ln(P(t) / P(t-1))$, where P is the closing price. Log returns are very close to ordinary percentage changes for small moves and are easier to work with mathematically.
2. **Summary statistics.** I computed the mean, standard deviation, skewness and excess kurtosis. For a normal distribution, skewness is 0 and excess kurtosis is 0. A positive excess kurtosis means heavier tails than normal.
3. **Histogram against a normal curve.** I plotted the real returns next to a normal curve with the same mean and standard deviation.
4. **QQ plot.** I plotted the quantiles of the real returns against the quantiles of a normal distribution. Points that follow the straight line indicate normality, and departures at the ends indicate heavier or lighter tails.
5. **Extreme-day count.** I standardised each return into a z-score, (return − mean) / standard deviation, and counted days beyond 3 and 4 standard deviations. Under normality, about 0.27% of days should fall beyond 3 and about 0.0063% beyond 4.
6. **Jarque–Bera test.** This test combines skewness and kurtosis into one statistic. A very small p-value means the data is very unlikely to have come from a normal distribution.

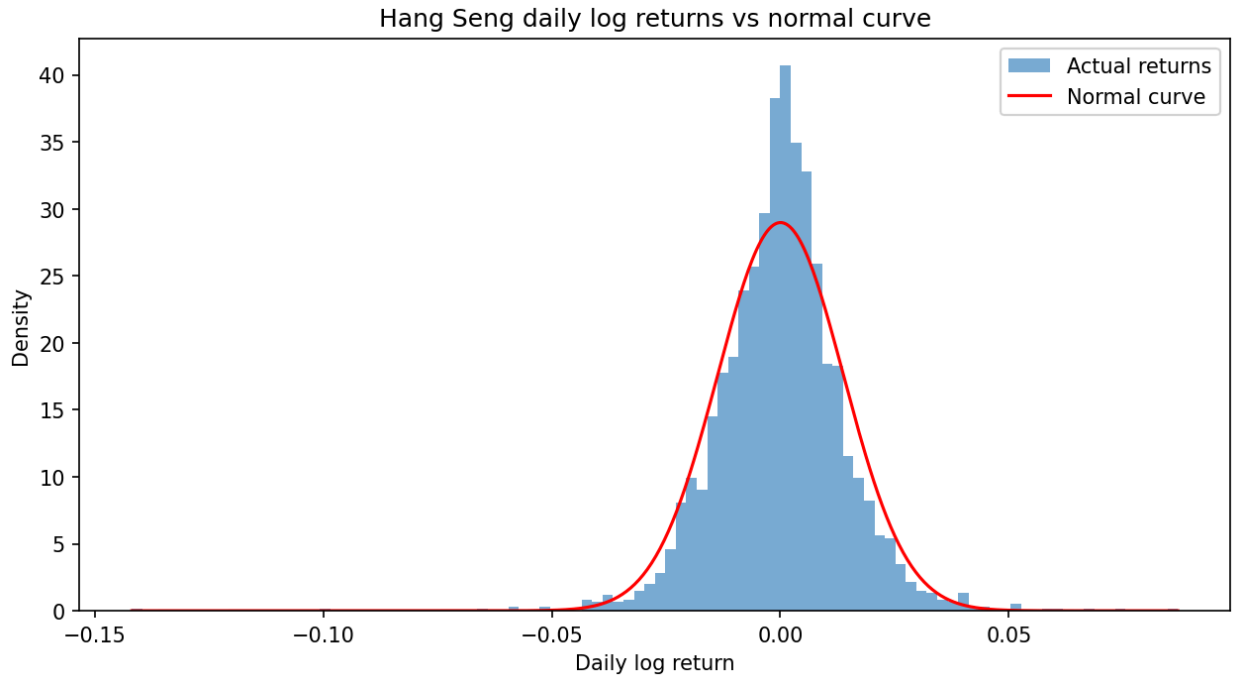
All analysis was done in Python (Google Colab) with pandas, NumPy, SciPy and Matplotlib.

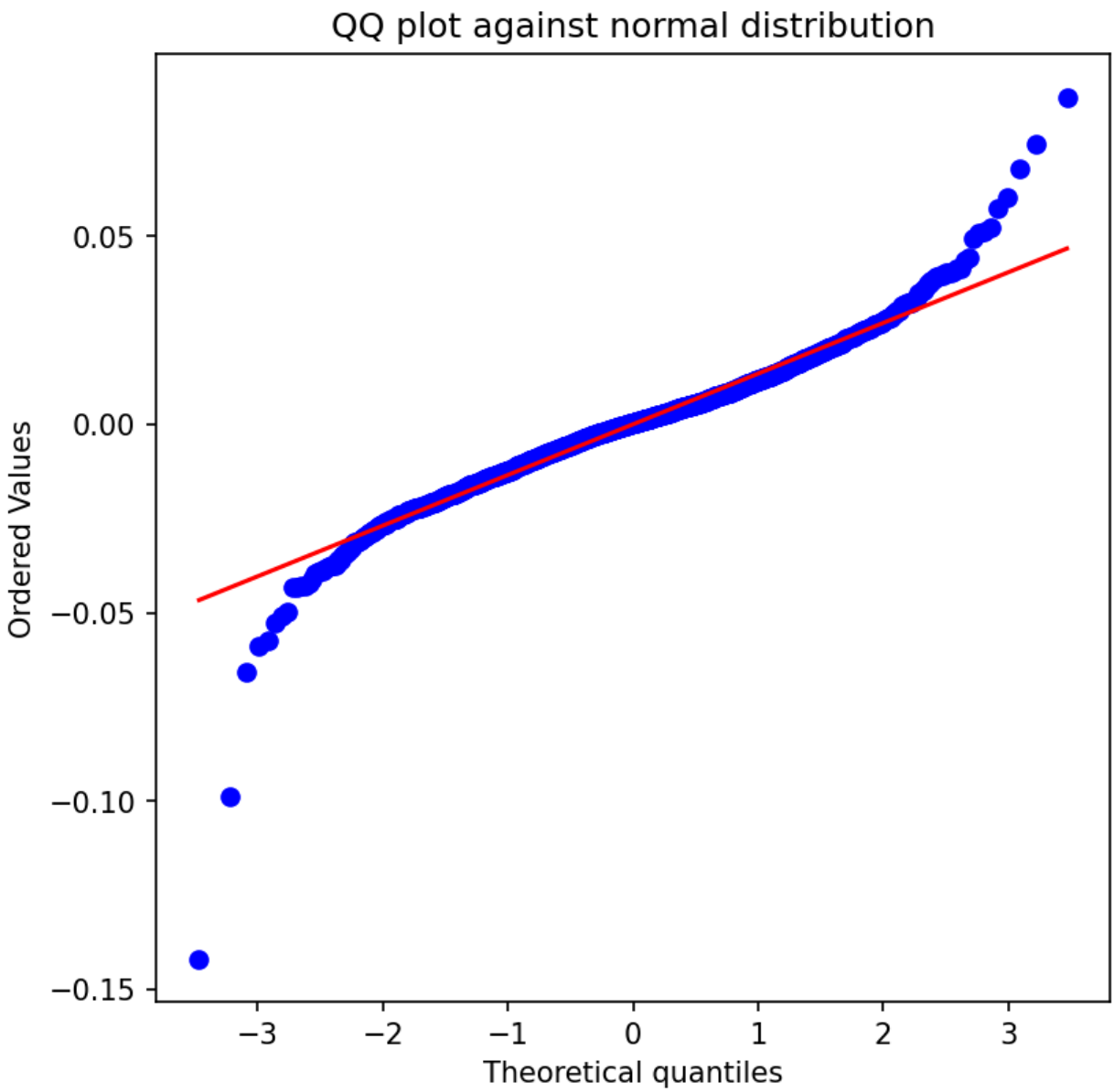
4. Results (Part 1)

Measure	**Value**
Daily Returns (observational)	2,653
Mean daily log return	0.00005
Standard deviation (daily)	0.01377

skewness	-0.371
Excess kurtosis	7.154
Days beyond 3 SD	24 (expected under normality: 7.2)
Days beyond 4 SD	10 (expected under normality: 0.17)
Jarque-Bera statistic	5,718.48
Jarque-Bera p-value	<0.001


Hang Seng Index closing price
Index level
32500
30000
27500
25000
22500
20000
17500
15000
2016
2018
2020
2022
2024
2026
Date


Hang Seng daily log returns vs normal curve
Actual returns
Normal curve
Density
40
35
30
25
20
15
10
5
0
−0.15
−0.10
−0.05
0.00
0.05
Daily log return



In Figure 2, the real returns are more concentrated around the centre and carry more weight in the far tails than the normal curve. In Figure 3, the points leave the straight line at both ends, which is the pattern expected when tails are heavier than normal.

The most extreme days

All ten days beyond four standard deviations are listed below. This matches the count of 10 in the table above. Percentages are ordinary one-day changes, converted from the log returns.

Largest falls

Date	Log Return	Change (%)
2025-04-07	-0.141833	-13.2
2024-10-08	-0.098793	-9.4
2022-10-24	-0.065673	-6.4
2022-03-15	-0.058867	-5.7
2020-05-22	-0.057202	-5.6

Largest rises

Date	Log Return	Change (%)
2022-03-16	0.086928	9.1
2022-11-11	0.074548	7.7
2022-03-17	0.068012	7.0
2024-10-02	0.060143	6.2
2022-10-05	0.057368	5.9

5. Interpretation

Returns are far from normal in the tails. An excess kurtosis of 7.15 means the tails are much heavier than a normal distribution allows. The extreme-day counts make this concrete: days beyond three standard deviations occurred about 3.3 times as often as normality predicts (24 against 7.2), and days beyond four standard deviations occurred about 59 times as often (10 against 0.17). Under a normal distribution, a four-standard-deviation day would be expected roughly once in 60 years of trading, and this sample contains ten of them. The Jarque–Bera result ($p < 0.001$) agrees.

Large losses were somewhat more common than large gains. The skewness of −0.37 shows a lean towards the downside. The single worst day (−13.2% on 7 April 2025) is larger in magnitude than the best day (+9.1%).

Extreme days cluster in time. Four of the five best days and two of the five worst fall in 2022, and the other extreme days also come in groups, such as the two consecutive large rises on 16

and 17 March 2022. Big falls and big rises tend to arrive in the same turbulent periods, with calmer stretches in between. A normal model treats every day as an independent draw with the same spread, so it cannot reproduce this. I noticed this pattern by inspecting the dates and did not test it formally.

Extreme days have possible connections to market news. The two largest falls I checked followed a major trade-policy announcement and a disappointment over expected economic stimulus. Sudden news of this kind is not something a smooth bell curve is designed to represent.

Consequence for risk models. A risk model that assumes normal returns would treat days like 7 April 2025 as practically impossible. Since several such days occurred in about ten years, such a model would understate the likelihood of large losses.

6. Part 2: Pricing an option by simulation

What the option is

A European call option gives its holder the right, but not the obligation, to buy the index at a fixed price (the strike) on a fixed future date (the expiry). If the index finishes above the strike, the holder gains the difference. If it finishes below, the option pays nothing.

Method

1. **Simulate index levels.** I modelled the index with geometric Brownian motion, the standard model behind Black–Scholes. The simulated index level at expiry is $S(T) = S(0) \times \exp[(r - \sigma^2/2)T + \sigma\sqrt{T} \times Z]$, where Z is a random draw from a standard normal distribution.
2. **Compute payoffs.** For each simulated path, the payoff is $\max(S(T) - K, 0)$.
3. **Average and discount.** The option price is the average payoff multiplied by $\exp(-rT)$.
4. **Check against Black–Scholes.** I compared the result with the closed-form Black–Scholes price for the same inputs, as a test that the simulation is correct.

Assumptions

Input	Value
Starting index level (S0)	24186.83 (latest close in the data)
Strike (K)	24186.83(at the money)

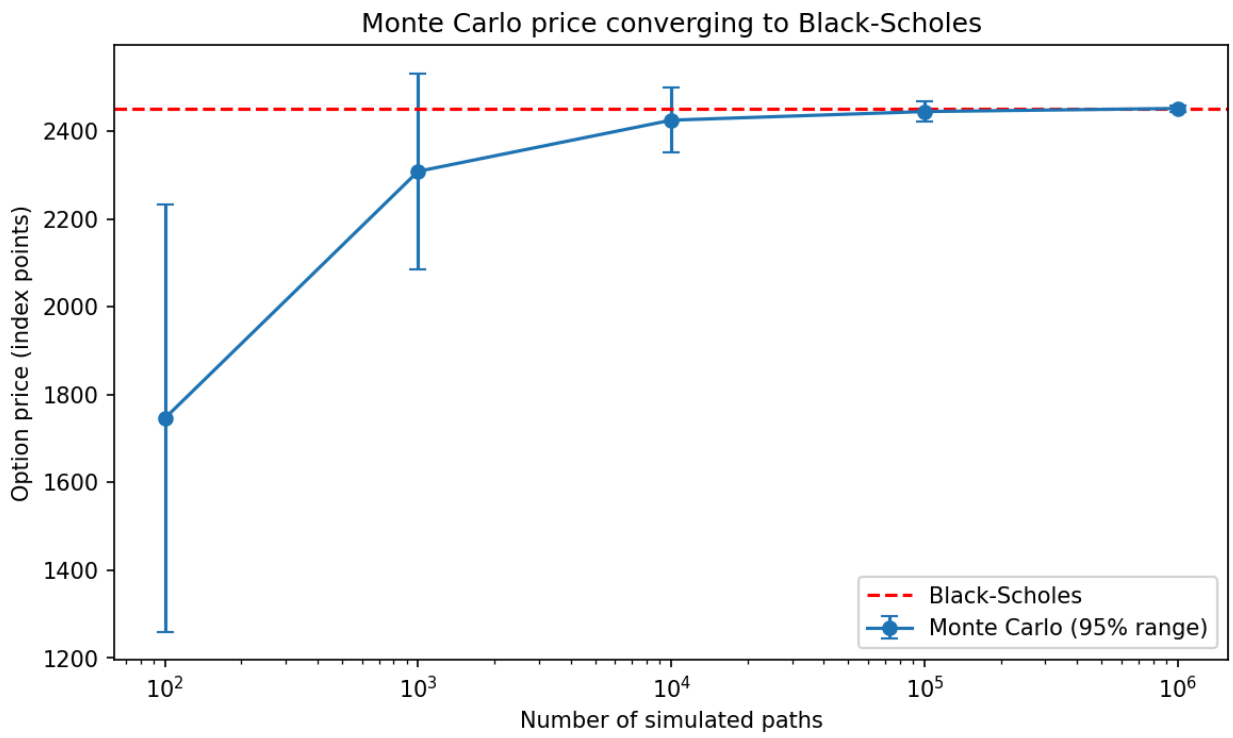
Time to expiry (T)	1 year
Risk free rate r	3% per year (an assumption, not market data)
Volatility σ	0.2186

Prices are in index points.

Results

The Black–Scholes price is **2,450.58** index points.

Paths	Monte Carlo Price	Standard error	Absolute difference
100	1746.02	247.91	704.56
1,000	2308.22	113.91	142.37
10,000	2424.68	37.62	25.91
100,000	2444.28	11.91	6.31
1,000,000	2451.35	3.75	0.76



The simulation converges to the Black–Scholes price. With 1,000,000 paths the difference is 0.76 index points, about 0.03%. The standard error falls roughly in proportion to one over the square root of the number of paths: going from 10,000 to 1,000,000 paths (100 times more) reduced it from 37.62 to 3.75, about a factor of 10. For 1,000 paths and above, the Black–Scholes price lies inside the Monte Carlo 95% range. With only 100 paths it does not, which shows how unreliable very small simulations are.

The agreement confirms that the simulation is implemented correctly.

How Part 2 connects to Part 1

Both the simulation and the Black–Scholes formula assume that returns are normally distributed. Part 1 shows that this assumption does not hold for the Hang Seng Index. The prices in Part 2 are therefore model prices, not market prices, and they are only as realistic as the normality assumption behind them.

7. Limitations

- **One index, one period.** The results describe the Hang Seng Index over roughly ten and a half years. They should not be generalised to other markets or periods without testing.

- **Sensitivity of the test.** With more than 2,600 observations the Jarque–Bera test can reject normality for small deviations. The conclusion here rests more on the size of the departure (excess kurtosis above 7 and the extreme-day counts) than on the p-value alone.
- **Z-scores use the sample's own mean and standard deviation,** which include the extreme days themselves.
- **Volatility clustering was observed, not tested.** I inferred it from the dates of extreme days and did not run a formal test.
- **Option price is a model price.** It uses a constant volatility estimated from historical data, an assumed 3% interest rate, no dividends, and normally distributed returns. Real option prices are set by the market and reflect expected future volatility.
- **Effect of fat tails on this option.** An option struck at the current index level is less sensitive to tail behaviour than options far from the current level, and the tail effect matters most for far out-of-the-money options and for measuring crash risk. This project does not quantify the pricing effect of fat tails.

8. Conclusion and possible extensions

Hang Seng Index daily returns are not normally distributed: they have much heavier tails than a normal distribution and a lean towards large losses, and extreme days occur far more often than normality predicts. A Monte Carlo option pricer built on the normal assumption matches Black–Scholes closely, so the method works, but the assumption behind it is contradicted by the data.

Natural next steps:

1. **Model changing volatility.** Fit a GARCH model to capture the clustering of calm and turbulent periods, and test whether it forecasts volatility better than a simple rolling average.
2. **Use heavier-tailed returns.** Repeat the simulation with a distribution such as the Student's t, and compare option prices with the normal case, especially for options far from the current level.
3. **Test other markets.** Repeat the analysis on other indices to see whether the pattern is specific to Hong Kong.

About this project

I completed this project in October 2026 as self-directed learning in Python and quantitative finance. The full code, charts and results files are available at https://github.com/farihatasnim04/hang_seng_returns_analysis.git .

References

- AFP (7 April 2025). “Hong Kong stocks plunge on worst day since 1997.”
- Associated Press (8 October 2024). “Hong Kong’s Hang Seng index plunges 9.5% as investors dump shares after recent sharp gains.”
- Yahoo Finance, Hang Seng Index (^HSI) historical data, accessed with the yfinance Python package.